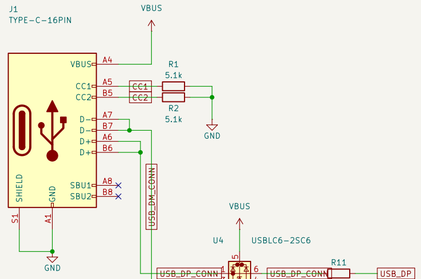
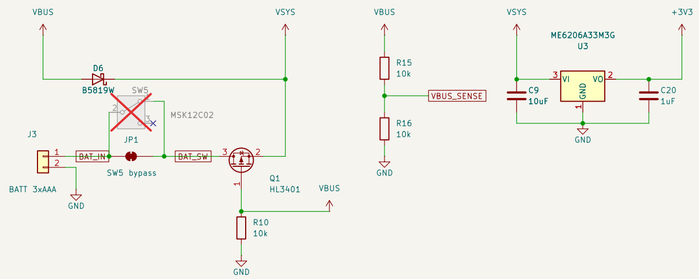
INTENT 2026 hardware badge — generated by 03-kicad/scripts/gen_sch.py from design.py (edit design.py, not this file).

USB-C (USB 2.0 full speed, device)



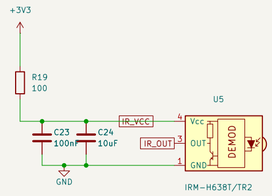
5.1k on CC1/CC2 = UFP (device). SBU pins unused.
USBLC6-2 ESD array in line, 27R series resistors
per the RP2040 hardware design guide.

POWER USB or 3xAAA -> VSYS -> 3V3 LDO



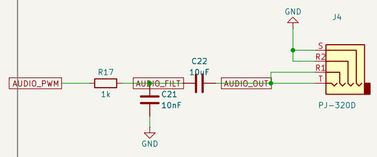
VSYS = 4.65 V on USB (after D6), 3.0–4.8 V on battery. With both present VBUS holds Q1 off, D6 blocks back-feed.
SW5 is DNP (the switch is on the battery box); JP1 bridges it. Cut JP1 only if SW5 is fitted.
RGB LEDs run from VSYS, everything else from 3V3 (ME6206, 300 mA).
R15/R16: VBUS_SENSE = 2.5 V on USB, 0 V on battery (GPIO29 / ADC3) so firmware can dim and sleep.

IR RECEIVER 38 kHz



Everlight IRM-H638T, 38 kHz carrier.
Open-collector out, internal pull-up, active low.
R19 + C24 = datasheet RC supply filter.
Pinout: 1,2 = GND, 3 = OUT, 4 = VCC (verified).
Keep >= 10 mm from the RGB LEDs on the board.

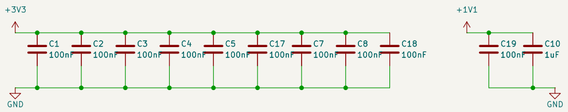
AUDIO 3.5 mm jack

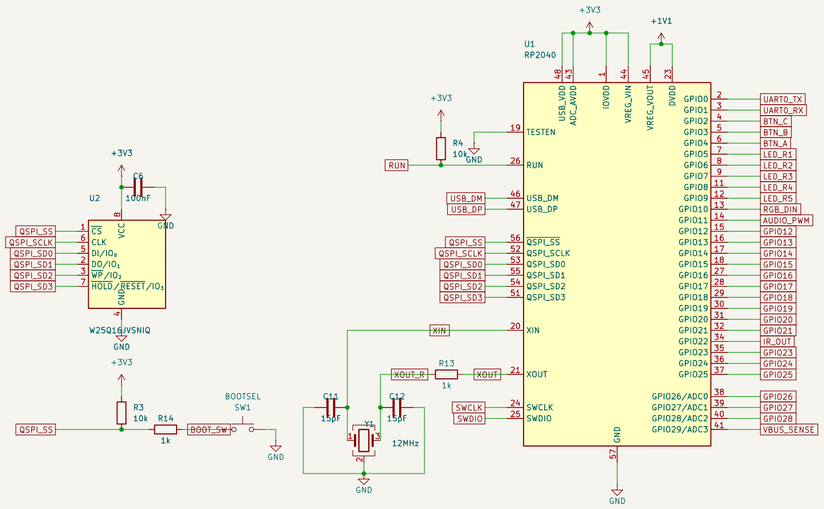


GPIO11 PWM (>= 100 kHz carrier) -> 1k / 10n low-pass (fc ~ 16 kHz)
-> 10 uF DC block -> tip + ring 1 (mono on both ears). Sleeve + ring 2 = GND.
Level into 32 ohm headphones ~ 0.1 Vpp (1k source). Detect-switch pads unused.
PJ-320D (C431535) is SMD: pads 1 = S, 2 = R2, 3 = R1, 4 = T.

RP2040 dual Cortex-M0+, 2 MB QSPI flash, 12 MHz crystal

Decoupling: one 100 nF per RP2040 supply pin (6x IOVDD, USB_VDD, ADC_AVDD, VREG_VIN); 1 uF + 100 nF on the 1V1 core rail.

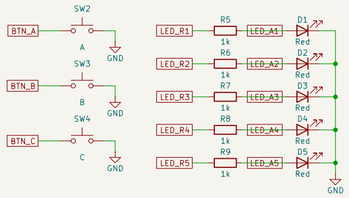




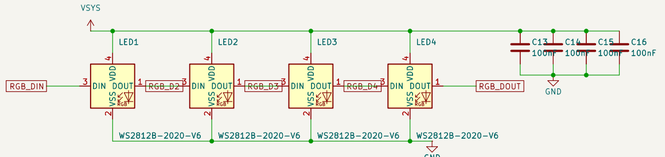
BOOTSEL: hold SW1 while plugging USB in -> UF2 bootloader. R14 protects the QSPI_SS driver.
Crystal 12 MHz: CL = 20 pF part with 15 pF loads (Pico values, ~ +50 ppm, fine for USB). R13 limits drive.
TESTEN tied to GND. RUN has a 10k pull-up and is on J2 for an external reset / debugger.

GPIO: 0/1 UART0 -> TP1/TP2; 4/3/2 = keys A/B/C;
5–9 red LEDs; 10 WS2812 data; 11 audio PWM;
22 IR receiver; 29 VBUS sense (ADC3);
12–21, 23–28 free on J2 (SPI1, I2C0, UART1, ADC0–2).

USER INTERFACE keys, credit LEDs, RGB shelf

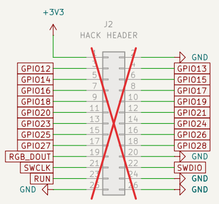


Keys A/B/C: active low, internal pull-ups, debounce in firmware.
Credit display: 5 red LEDs, 1k series -> ~1.5 mA each (dim by design).



WS2812B-2020-V6 x4 on VSYS (not 3V3), 100 nF each. Data from GPIO10, chain out to J2 pin 19.
DIN is driven at 3.3 V against VIH = 0.7 x VDD: marginal on paper, works in practice.

HACK HEADER J2 (DNP), test points, lanyard holes









PWR_FLAGs: nets fed only through passive pins.

2x13, 2.54 mm, not fitted — players solder their own pins. 3V3 logic only, no 5 V on the header.
Pin 1 = top-left square pad. GND on both end rows: a reversed ribbon shorts 3V3 to GND at worst (LDO limits).
16 free GPIO + SWD + RUN + RGB chain out. TP1–TP3: UART0 console 115200 8N1. H1/H2: 4.0 mm plated lanyard holes.

Generated by scripts/gen_sch.py — edit design.py, not this file
RP2040 badge, 56x100 mm vending-machine outline
INTENT 2026 — by Palo Alto Networks

Sheet: /
File: intent-2026-badge.kicad_sch

Title: INTENT 2026 Hardware Badge

Size: A2	Date: 2026-10-11	**Rev: 0.4**
KiCad E.D.A. 10.0.3		Id: 1/1